Ömer Gölcük 38101 **EE58003: VLSI Design I Lab 4 Report**

Introduction

In this laboratory, the complete physical design flow of a Multiply–Accumulate (MAC) unit is carried out using a hierarchical place-and-route (P&R) methodology in Cadence Innovus. The objective of this lab is to transform a synthesized gate-level netlist into a manufacturable layout while satisfying physical, timing, and power-related constraints. The design flow begins with floorplanning, where the core area, aspect ratio, and placement boundaries are defined. This is followed by I/O ring integration and power planning to ensure robust power delivery throughout the design. Standard cell placement is then performed, after which clock tree synthesis (CTS) is applied to distribute the clock signal while controlling skew and insertion delay. Global and detailed routing stages are subsequently executed to complete all signal interconnections. After the P&R flow is completed, the resulting layout is validated through signoff checks including design rule checking (DRC), connectivity verification, and antenna rule checks to ensure physical correctness and manufacturability. The final GDSII layout file is then analyzed using Cadence Virtuoso to visually inspect the physical implementation and confirm the integrity of the layout.

Design Overview and Setup

The design implemented in this laboratory is a Multiply–Accumulate (MAC) unit, which constitutes a fundamental building block in many digital signal processing and machine learning applications. The MAC unit performs multiplication of input operands followed by accumulation of the result, and its regular structure makes it well suited for evaluating physical design methodologies. A hierarchical design approach is adopted throughout the place-and-route flow. Instead of flattening the entire design, the MAC unit is composed of multiple submodules, allowing improved scalability, better manageability of physical constraints, and more efficient optimization during placement and routing. This approach also reflects industry-standard physical design practices for medium- to large-scale designs.

The physical design process is carried out using Cadence Innovus, which is employed for floorplanning, power planning, standard cell placement, clock tree synthesis, and routing. The input to Innovus consists of a synthesized gate-level netlist of the MAC unit, associated timing constraints, and technology-specific libraries. Standard cell libraries and physical abstract views (LEF) are used to define cell geometries and routing rules, while technology files specify layer definitions and design rules. Timing constraints are provided to guide placement and clock tree synthesis, ensuring that the design meets its target clock requirements. Additionally, power and ground nets are explicitly defined at the beginning of the flow to enable proper power planning and verification. All design steps are performed in a consistent hierarchical environment, allowing each stage of the P&R flow to build upon the results of the previous stage.

Floorplanning

The floorplanning stage establishes the physical foundation of the MAC unit by defining the die and core regions, placement boundaries, and overall layout organization. At this stage, the core area is sized to accommodate standard cells while targeting a reasonable utilization to balance routability and area efficiency. The aspect ratio is selected to minimize routing congestion and to align with the hierarchical structure of the design. I/O pads are placed around the periphery of the die, enclosing the core region. This configuration enables clear separation between core logic and external interfaces, simplifying power distribution and signal routing. The hierarchical nature of the design is preserved during floorplanning, ensuring that submodules are placed consistently within the core area.

I/O Ring Integration

After defining the core region during floorplanning, the I/O ring is integrated to provide a structured interface between the internal logic of the MAC unit and the external environment. The I/O ring surrounds the core area and contains the pad cells required for signal input/output as well as power and ground connections. The placement of I/O pads is performed in a manner consistent with the signal directions and functional grouping of the design. This organization simplifies routing between the core logic and the pads while maintaining a clear separation between internal standard cells and peripheral circuitry. The I/O ring also serves as a physical boundary that encloses the core region and facilitates robust power delivery when combined with the power distribution network. At this stage, logical connectivity between core signals and corresponding I/O pads is established, but detailed signal routing is deferred to later stages of the P&R flow. Integrating the I/O ring early ensures that placement, clock tree synthesis, and routing stages are performed with accurate boundary conditions and realistic physical constraints. The correctness of the I/O ring integration is later verified implicitly through successful routing and connectivity checks, confirming that all external interfaces are properly connected without violations.

Power Planning

Following I/O ring integration, power planning is performed to establish a reliable power distribution network (PDN) for the MAC unit. The objective of this step is to ensure stable delivery of power and ground signals to all standard cells while minimizing voltage drop and improving overall power integrity. Power and ground rings are generated around the core region to provide a low-resistance path for VDD and VSS. These rings are implemented on higher metal layers to reduce resistive losses and to serve as robust sources for internal power distribution. The power ring configuration is chosen to fully enclose the core, enabling uniform power delivery from the I/O pads to the internal logic.

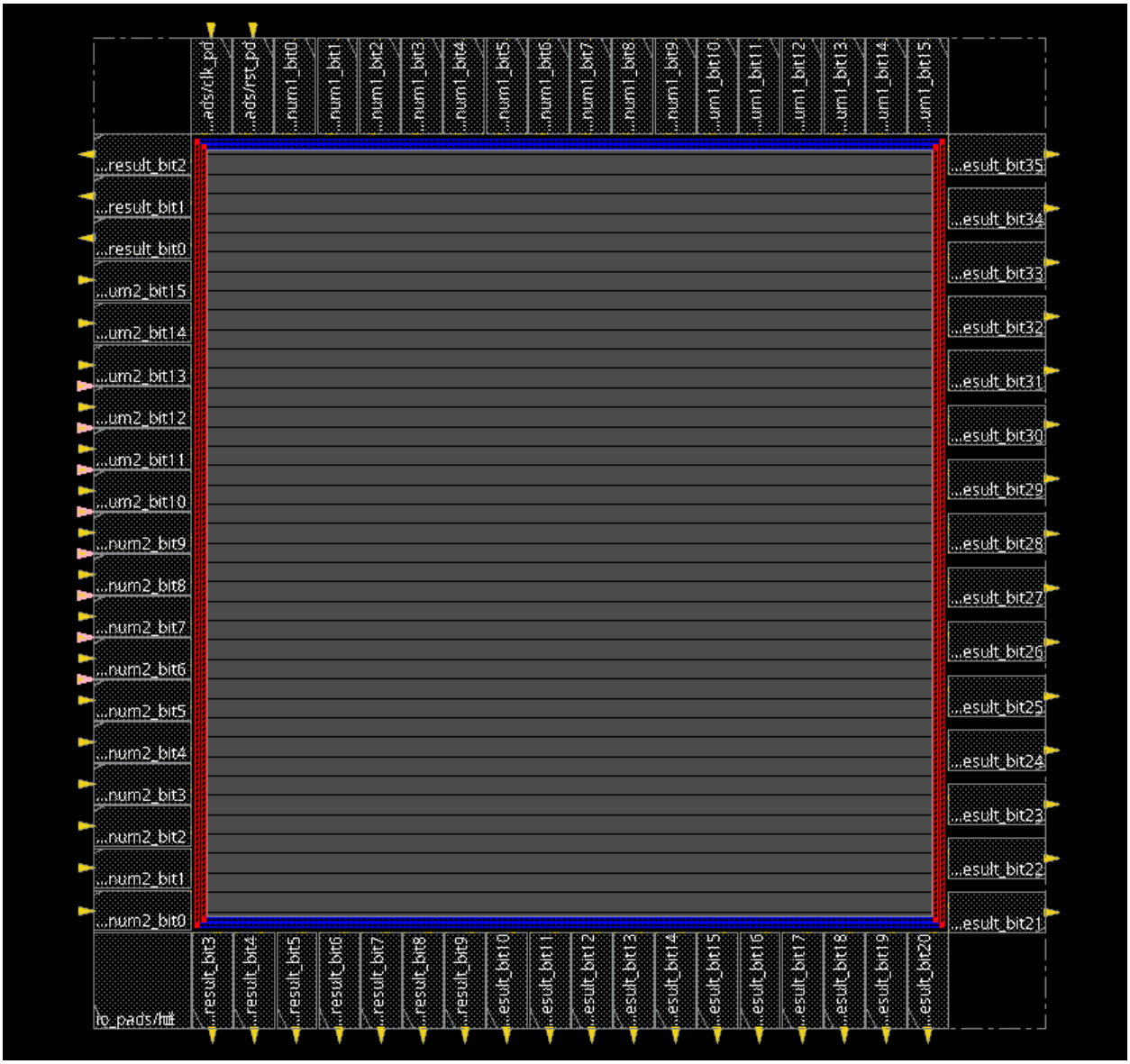

Figure 1: Power Ring Added to Layout

After the creation of the power rings, special routing is applied to connect the global power and ground nets throughout the design. This step ensures that all power pins of standard cells are properly connected to the PDN using non-default routing rules suitable for power delivery. Special routing is performed prior to standard signal routing to prevent power integrity issues and routing conflicts in later stages.

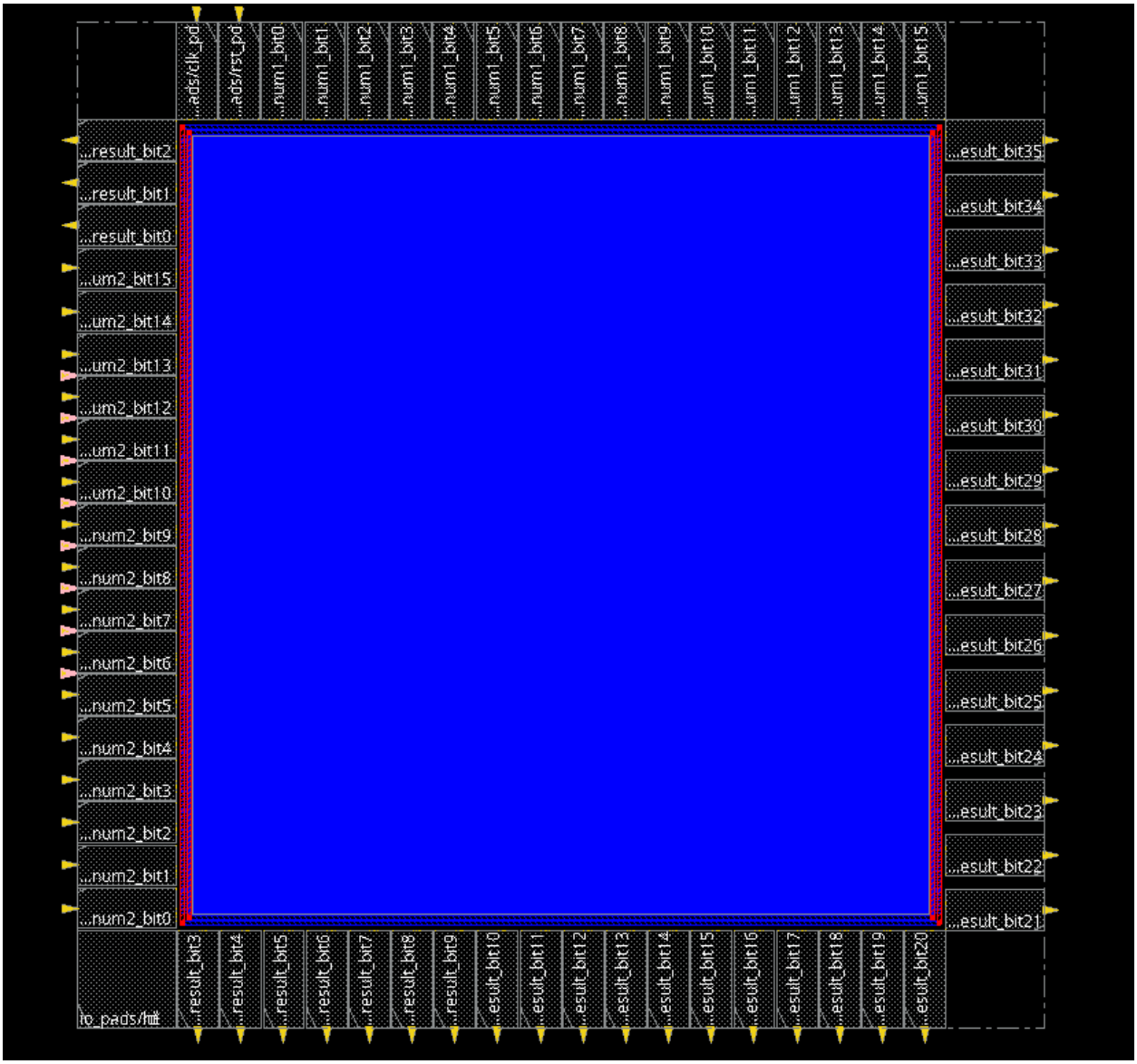

Figure 2: The Layout After Power Routing Completed

The successful completion of power planning provides a stable foundation for placement, clock tree synthesis, and routing, and is critical for achieving a physically correct and manufacturable layout.

Standard Cell Placement

After completing power planning, standard cell placement is performed to assign physical locations to all logic cells within the core region. This step aims to achieve an efficient distribution of cells that balances area utilization, timing requirements, and routing feasibility.

The placement process is executed using the placeDesign command in Cadence Innovus. During this stage, standard cells are positioned within the predefined core area while respecting placement constraints, power connections, and design boundaries established in earlier steps. The initial placement targets a reasonable density to avoid excessive congestion in later routing stages.

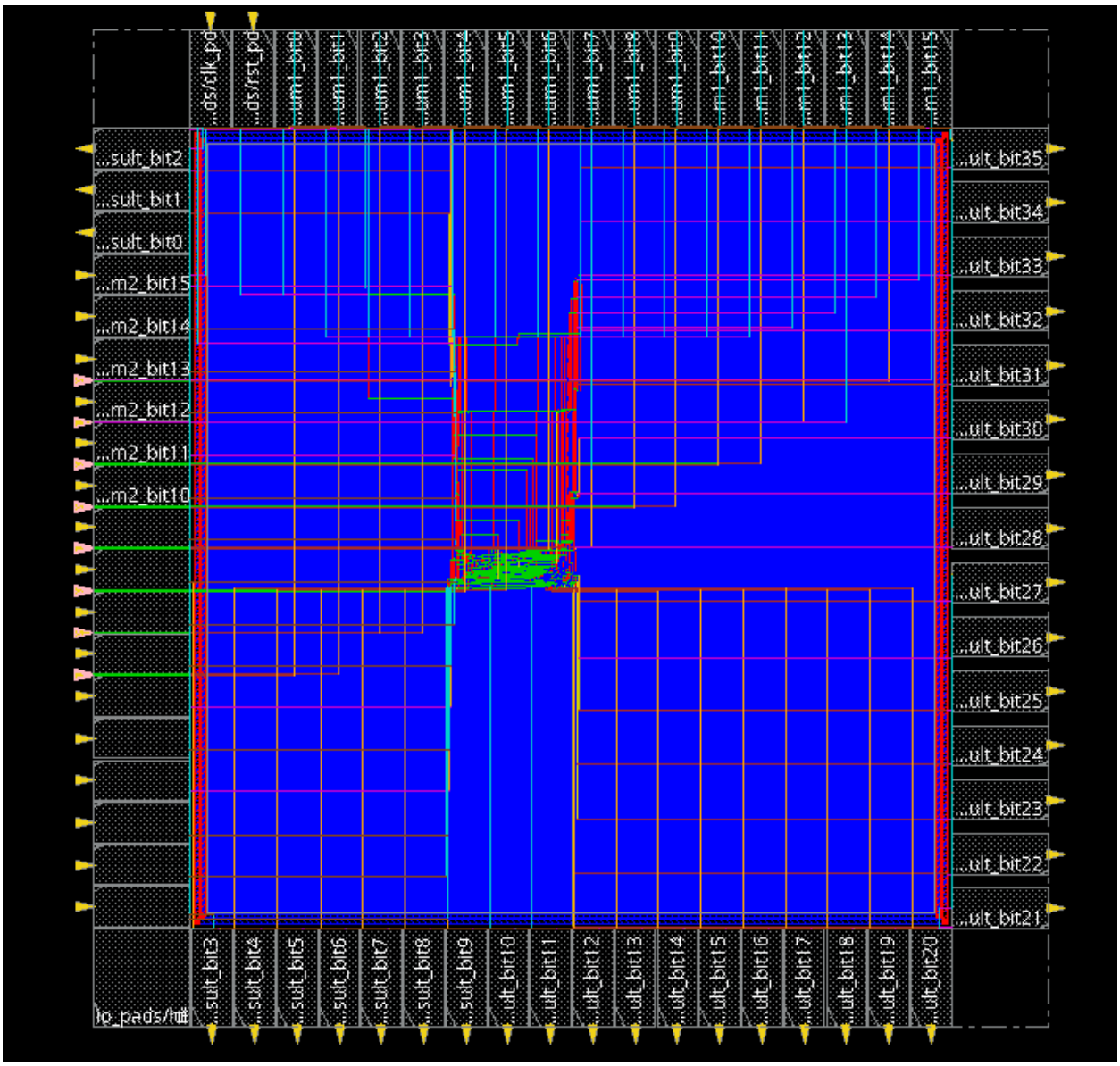

Figure 3: Layout after placeDesign

Following initial placement, pre-clock tree synthesis optimization is applied using the optDesign -preCTS command. This optimization step improves timing slack and placement quality by refining cell locations and resolving early congestion or critical path issues before clock tree synthesis is introduced. Performing this optimization prior to CTS helps reduce the complexity of clock distribution and improves overall timing closure.

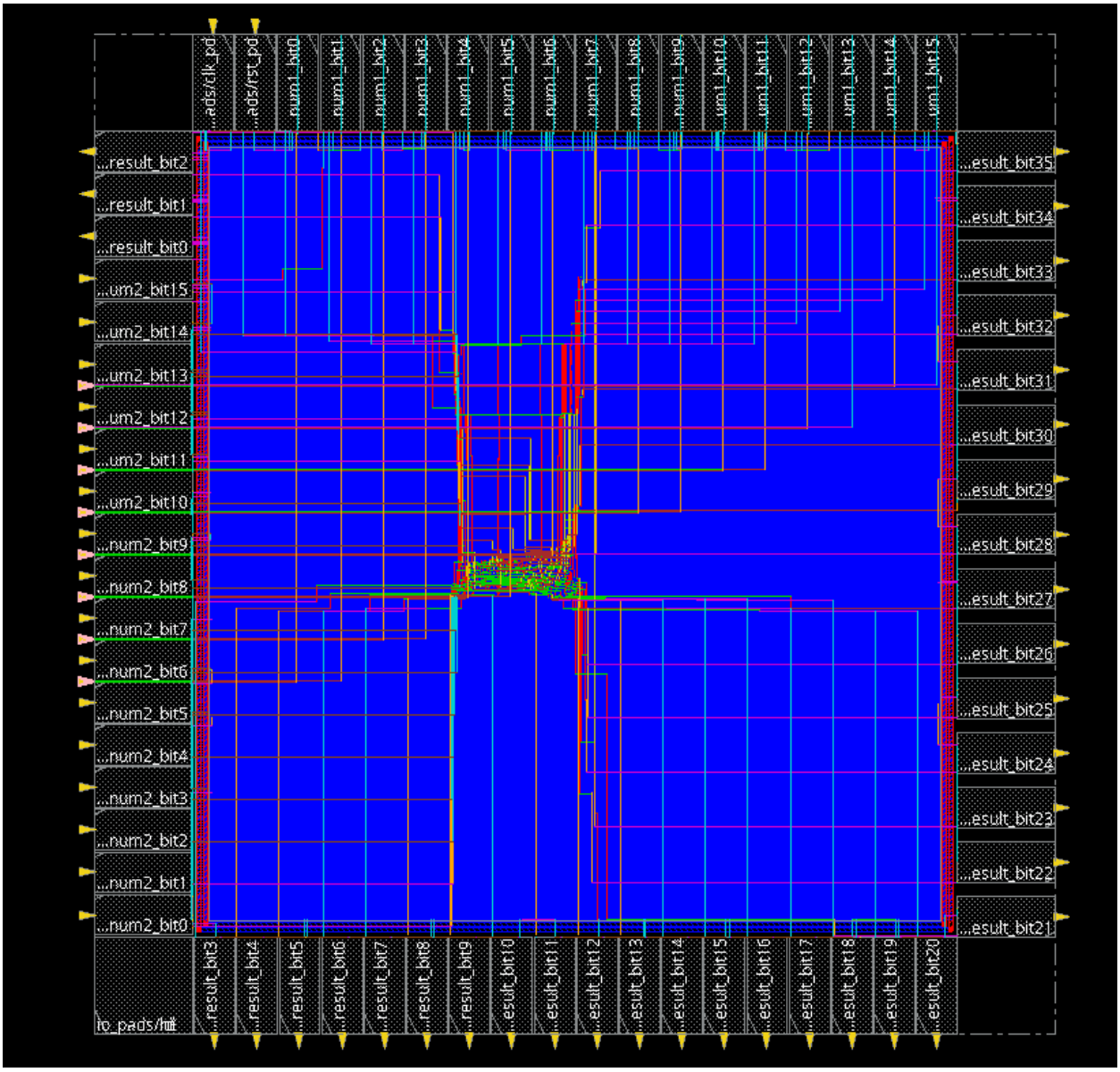

Figure 4: Layout after optDesign -preCTS

Clock Tree Synthesis (CTS)

After completing standard cell placement and pre-CTS optimization, clock tree synthesis (CTS) is performed to distribute the clock signal to all sequential elements in the design while minimizing skew and controlling insertion delay. CTS is a critical step, as improper clock distribution can lead to setup or hold time violations even if data paths are well optimized.

```
------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------

Setup views included:
 TC_AV

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):| -0.003  | -0.003  |  7.059  |
|           TNS (ns):| -0.004  | -0.004  |  0.000  |
|    Violating Paths:|    2    |    2    |    0    |
|          All Paths:|   84    |   68    |   84    |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |     52 (52)      |
|   max_fanout   |      1 (1)       |    -15     |      2 (2)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 0.662%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------
Reported timing to dir ./timingReports
Total CPU time: 1.09 sec
Total Real time: 3.0 sec
Total Memory Usage: 1562.277344 Mbytes
0
innovus 4> innovus 4>
```

Figure 5:Output of timeDesign -preCTS

Clock constraints defined during the setup phase are used to guide the synthesis of the clock network. The CTS process inserts clock buffers and constructs a balanced clock tree structure to ensure uniform clock arrival times across all registers. Special care is taken to reduce clock skew and to limit excessive clock latency that could negatively impact timing.

Following clock tree construction, post-CTS optimization is applied using the optDesign -postCTS command. This step refines both clock and data paths to address timing degradations introduced

by clock tree insertion. Timing analysis is then performed using timeDesign -postCTS to verify that timing requirements are satisfied after CTS.

```
------------------------------------------------------------
     optDesign Final Summary
------------------------------------------------------------

Setup views included:
 TC_AV

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.005  |  0.005  |  7.050  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   84    |   68    |   84    |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |     52 (52)      |
|   max_fanout   |      1 (1)       |    -15     |      2 (2)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 0.668%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------
**optDesign ... cpu = 0:01:07, real = 0:01:08, mem = 1395.7M, totSessionCpu=0:10:22 **
*** Finished optDesign ***
Info: Destroy the CCOpt slew target map.
Removing temporary dont_use automatically set for cells with technology sites with no row.
0
innovus 6> innovus 6>
```

Figure 6:Output of optDesign -postCTS

```
# Design Name: MAC_TOP
# Design Mode: 90nm
# Analysis Mode: MMMC Non-OCV
# Parasitics Mode: No SPEF/RCDB
# Signoff Settings: SI Off
#################################################################################
Calculate delays in Single mode...
Start delay calculation (fullDC) (1 T). (MEM=1556.57)
Total number of fetched objects 2509
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=1583.78 CPU=0:00:00.6 REAL=0:00:01.0)
End delay calculation (fullDC). (MEM=1583.78 CPU=0:00:00.8 REAL=0:00:01.0)
*** Done Building Timing Graph (cpu=0:00:00.9 real=0:00:01.0 totSessionCpu=0:08:39 mem=1583.8M)

------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------

Setup views included:
 TC_AV

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):| -0.053  | -0.053  |  7.058  |
|           TNS (ns):| -0.104  | -0.104  |  0.000  |
|    Violating Paths:|    2    |    2    |    0    |
|          All Paths:|   84    |   68    |   84    |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |     52 (52)      |
|   max_fanout   |      1 (1)       |    -15     |      2 (2)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 0.662%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------
Reported timing to dir ./timingReports
Total CPU time: 1.86 sec
Total Real time: 3.0 sec
Total Memory Usage: 1550.035156 Mbytes
0
innovus 5> innovus 5>
```

Figure 7:Output of timeDesign -postCTS

The clock network is further inspected using the clock tree debugger, which provides a visual representation of the synthesized clock structure and confirms the balanced distribution of the clock signal throughout the design.

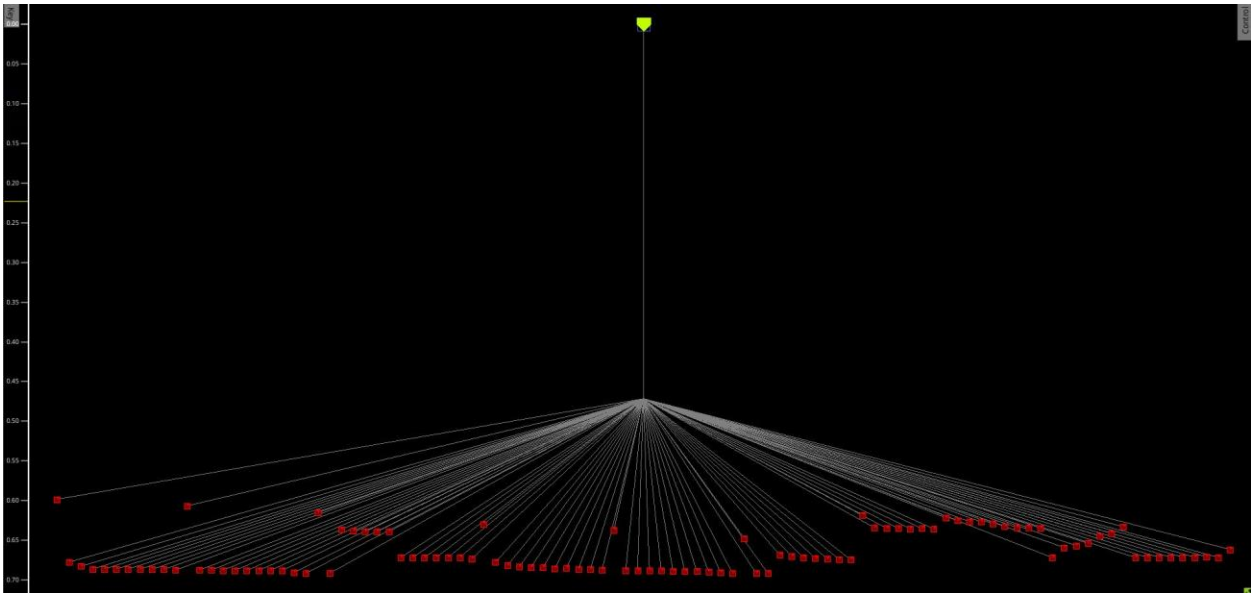

Figure 8:Output of the Clock Tree Debugger

Routing

After clock tree synthesis and post-CTS optimization, routing is performed to establish all signal interconnections within the design. This stage completes the physical connectivity of the MAC unit by routing both clock and data nets while complying with design rules and timing constraints.

```
------------------------------------------------------------
     optDesign Final SI Timing Summary
------------------------------------------------------------

Setup views included:
 TC_AV

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.031  |  0.031  |  6.994  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   84    |   68    |   84    |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |     52 (52)      |
|   max_fanout   |      1 (1)       |    -15     |      2 (2)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 0.668%
       (98.885% with Fillers)
Total number of glitch violations: 0
```

Figure 9: optDesign -postRoute

Routing is carried out in two phases: global routing and detailed routing. During global routing, approximate routing paths are determined to assess congestion and routing feasibility across the layout. This is followed by detailed routing, where exact wire geometries are assigned to specific metal layers while satisfying spacing, width, and via rules defined by the technology.

Following detailed routing, post-route optimization is applied using the optDesign -postRoute command to further improve timing and resolve any remaining routing-related issues. Timing analysis is then performed using timeDesign -postRoute to verify that all timing constraints are met after routing. These steps ensure that the design achieves timing closure under realistic post-routing conditions.

```
------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------

Setup views included:
 TC_AV

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.031  |  0.031  |  6.994  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   84    |   68    |   84    |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |     52 (52)      |
|   max_fanout   |      1 (1)       |    -15     |      2 (2)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 0.668%
       (98.885% with Fillers)
Total number of glitch violations: 0
------------------------------------------------------------
Reported timing to dir ./timingReports
Total CPU time: 4.3 sec
Total Real time: 6.0 sec
Total Memory Usage: 1648.597656 Mbytes
Reset AAE Options
0
```

Figure 10: timeDesign -postRoute

The routing process completes without violations, indicating that the earlier placement and power planning decisions successfully enabled a routable and physically correct design.

Post-Route Timing and Power Analysis

After routing is completed, post-route analysis is performed to evaluate the timing and power characteristics of the design under realistic physical conditions. Timing analysis is carried out using the timeDesign -postRoute command to verify that all timing constraints are satisfied after accounting for routing parasitics. In addition to timing, power analysis is conducted using the

report_power command. This report provides an estimate of the total power consumption of the MAC unit based on post-layout information. The reported power values reflect the combined effects of cell switching activity, clock network power, and routing parasitics, offering a realistic assessment of the design's power behavior.

```
Total Power
-----------------------------------------------------------------------------------------
Total Internal Power:       28.26222804            58.9150%
Total Switching Power:      19.65473738            40.9720%
Total Leakage Power:         0.05420913             0.1130%
Total Power:                47.97117404
-----------------------------------------------------------------------------------------


Group                           Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
Sequential                        0.1468     0.01464     3.3e-05      0.1615      0.3366
Macro                                  0           0           0           0           0
IO                                 27.42        19.1   0.0003478       46.52       96.97
Physical-Only                          0           0     0.05352     0.05352      0.1116
Combinational                     0.6955      0.5409   0.0003099       1.237       2.578
Clock (Combinational)                  0           0           0           0           0
Clock (Sequential)                     0           0           0           0           0
-----------------------------------------------------------------------------------------
Total                              28.26       19.65     0.05421       47.97         100
-----------------------------------------------------------------------------------------


Rail                  Voltage   Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
Default                   1.2      27.42        19.1   0.0003478       46.52       96.97
VDD                       1.2     0.8423      0.5556     0.05386       1.452       3.026


Clock                           Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
clk                               0.4824           0   4.268e-06      0.4824       1.006
-----------------------------------------------------------------------------------------
Total                             0.4824           0   4.268e-06      0.4824       1.006
-----------------------------------------------------------------------------------------
Clock: clk
Clock Period: 0.008000 usec
Clock Toggle Rate:   250.0000 Mhz
Clock Static Probability:  0.5000
```

Figure 11: Output of report_power

Signoff and Verification

After post-route timing and power analysis, the design undergoes final signoff and verification checks to ensure physical correctness and manufacturability. These checks are essential to confirm that the layout complies with technology design rules and that all electrical connections are properly established.

Design Rule Checking (DRC) is performed to verify compliance with technology-specific layout rules. The DRC results indicate the presence of a limited number of short, non-critical violations, which are acceptable within the scope of this laboratory as specified in the lab guidelines. These violations do not affect the overall physical correctness of the design or the validity of the place-and-route flow

```
 Verification Complete : 16 Viols.

Violation Summary By Layer and Type:

                 Short   Totals
       M3           10       10
       M5            4        4
       M6            1        1
       M7            1        1
       Totals       16       16

*** End Verify DRC (CPU: 0:00:06.4  ELAPSED TIME: 6.00  MEM: 0.0M) ***
```

Figure 12: Output of DRC check in Cadence Innovus showing a limited number of short, non-critical violations acceptable within the scope of the lab.

Connectivity checks are then conducted to confirm that all nets are correctly connected and that no unintended shorts or open circuits exist in the design. This step validates the logical consistency between the schematic-level netlist and the physical layout.

```
innovus 10> VERIFY_CONNECTIVITY use new engine.

******** Start: VERIFY CONNECTIVITY ********
Start Time: Fri Jan  9 01:10:34 2026

Design Name: MAC_TOP
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (1180.0000, 1228.6000)
Error Limit = 1000; Warning Limit = 50
Check all nets

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Fri Jan  9 01:10:38 2026
Time Elapsed: 0:00:04.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:03.2  MEM: 143.730M)
```

Figure 13:Output of Connectivity Checks

Finally, antenna checks are performed to identify and prevent antenna rule violations that could lead to gate oxide damage during fabrication. The absence of antenna violations indicates that the routing strategy is safe with respect to plasma-induced charging effects.

```
innovus 10>
******* START VERIFY ANTENNA ********
Report File: MAC_TOP.antenna.rpt
LEF Macro File: MAC_TOP.antenna.lef
Verification Complete: 0 Violations
******* DONE VERIFY ANTENNA ********
(CPU Time: 0:00:00.2  MEM: 0.000M)
```

Figure 14:Output of the Antenna Checks in Innovus

The successful completion of these checks confirms that the design is physically correct and ready for final layout generation.

Final Layout Overview in Cadence Virtuoso

After completing the place-and-route flow and signoff checks in Cadence Innovus, the final layout of the MAC unit is exported in GDSII format. This GDSII file is then imported into Cadence Virtuoso for visual inspection and analysis of the complete physical implementation.

Cadence Virtuoso provides a detailed view of the final layout, allowing examination of the overall chip structure, including the core region, I/O ring, power distribution network, clock routing, and signal interconnects. This step serves as a final sanity check to confirm that the exported layout accurately reflects the completed physical design.

The layout is first inspected with pad cells closed, providing a compact view of the overall design and its physical boundaries. This view allows verification of the I/O ring placement and the enclosure of the core region within the pad frame.

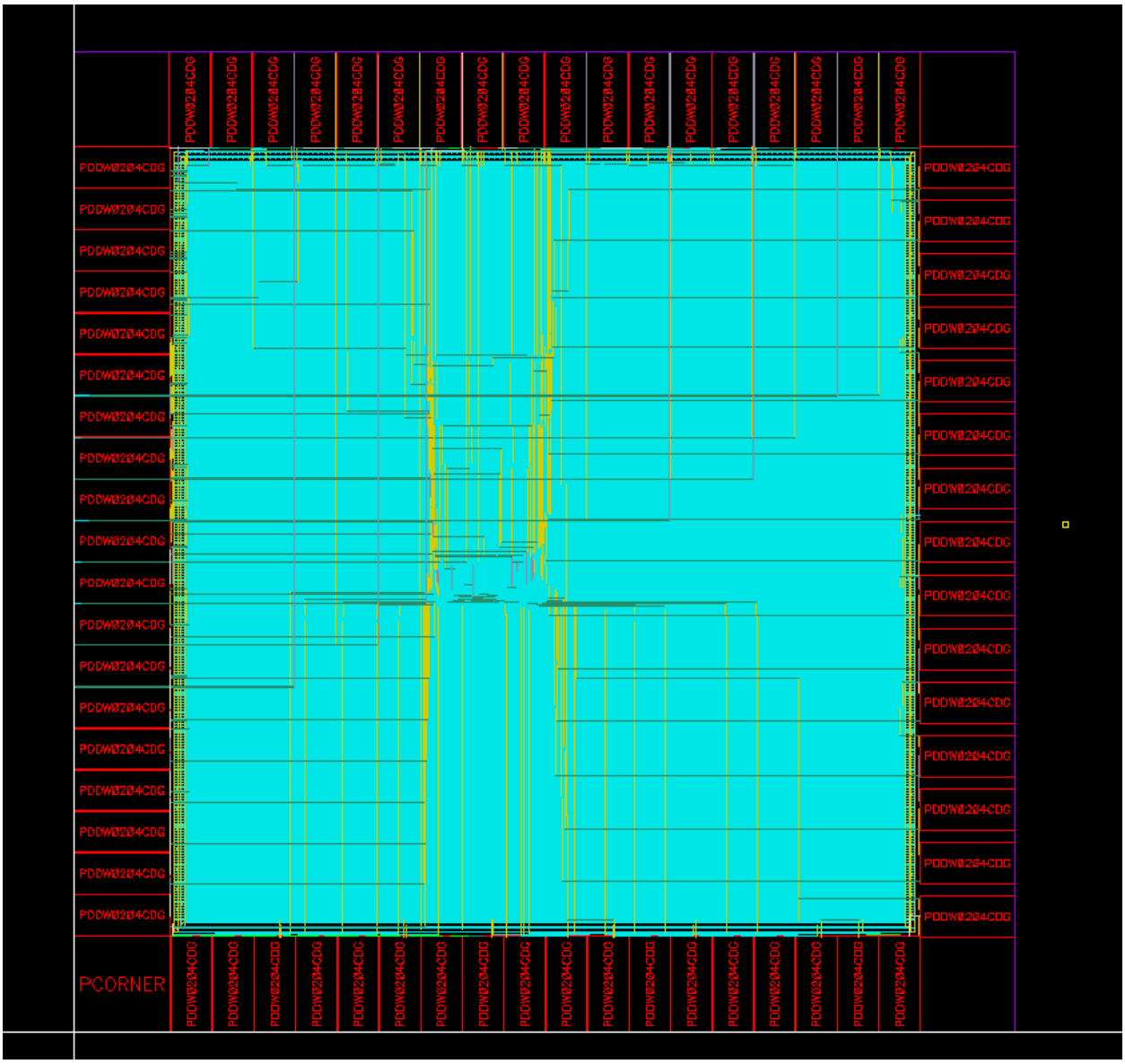

Figure 15:Layout of the MAC unit without SHIFT+F in virtuoso

Subsequently, the layout is examined using the *Shift+F* feature, which expands the hierarchical view and reveals the internal structure of the design. This enhanced visibility enables closer inspection of standard cell placement, routing density, and power grid organization within the core.

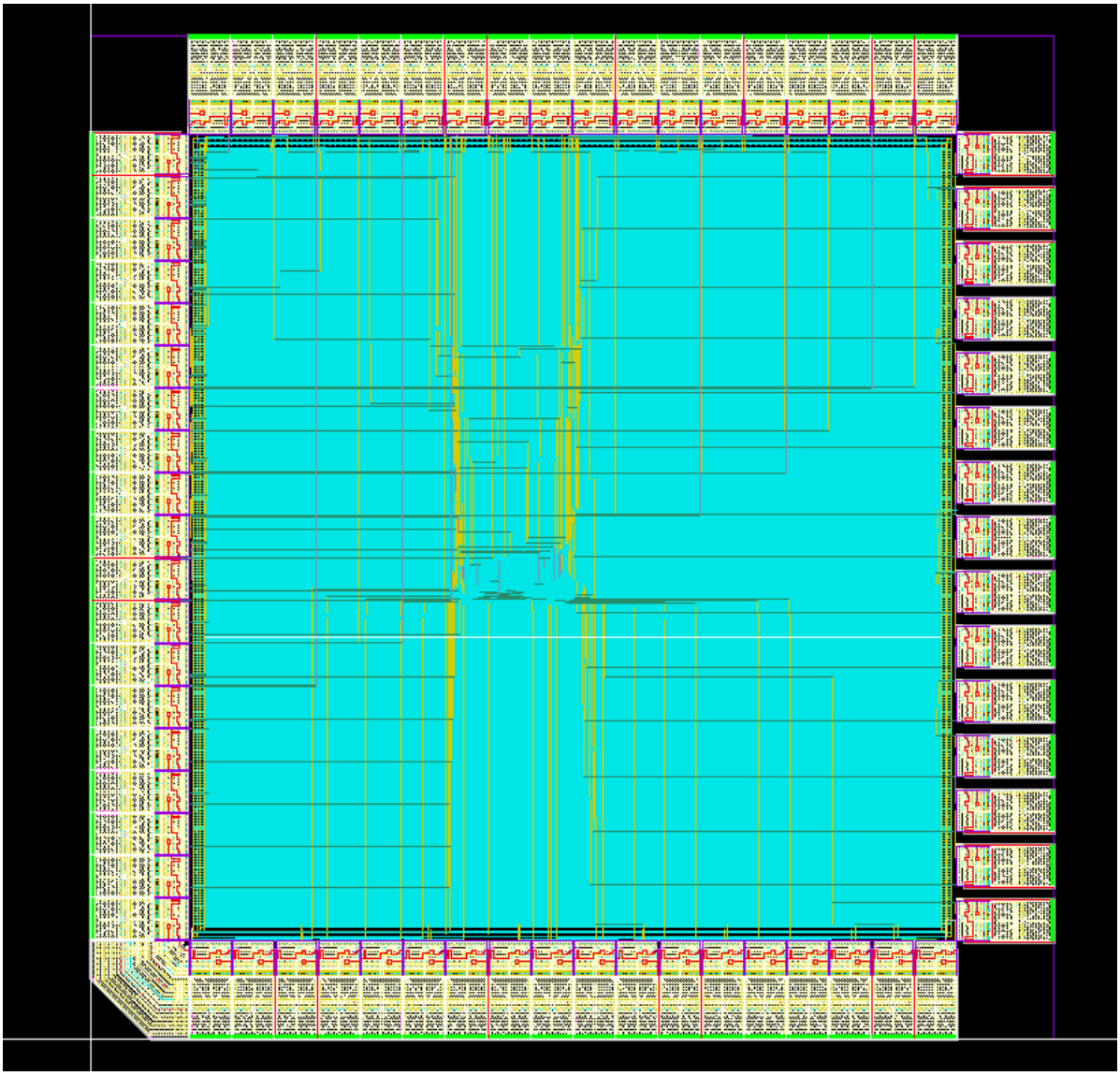

Figure 16:Layout of the design with shift + f

Since all DRC, connectivity, and antenna checks have already been successfully completed in Cadence Innovus, no additional physical verification is performed in Virtuoso.

Conclusion

In this laboratory, the complete physical design flow of a MAC unit was successfully implemented using a hierarchical place-and-route methodology in Cadence Innovus. The design progressed through all major physical design stages, including floorplanning, I/O ring integration, power planning, standard cell placement, clock tree synthesis, and routing.Post-route timing and power analyses confirmed that the design meets its timing requirements and operates within acceptable power limits under realistic physical conditions. Comprehensive signoff checks, including DRC (completed without critical violations) , connectivity, and antenna verification, were completed

without violations, ensuring that the layout is physically correct and manufacturable.The final GDSII layout was examined in Cadence Virtuoso, providing a complete visual confirmation of the post-layout physical implementation. Overall, this lab demonstrated a structured and industry-relevant physical design flow and highlighted the importance of each stage in achieving a correct and optimized VLSI layout.